Supporting Teachers' Data-Based Instruction in Early Writing: Tools, Learning, and Collaborative Support

Grant #R324A130144

Accomplishments

A. What are the major goals of the project?

As described in the performance agreement, the major goals of the project are to:

(1) Develop a professional development system (DBI-TLC) to support teachers' use of data-based instruction with students with or at risk for disabilities that affect early writing development, and
(2) Assess the feasibility and potential for efficacy of the intervention for improving student writing outcomes.

B. What was accomplished under these goals?

The status of the major activities of the project described in our Performance Agreement for Years 1 & Year 2 (September 1, 2013-August 31, 2015) are listed in Table 1, and described in more detail in the narrative below.

Table 1
Project Activities and Status

Project Activity (As listed in Performance Agreement)	Status
1. Develop and revise the DBI protocol (Manual with processes and tools).	Draft Completed (Yr 1); Revised (Yr 2)
2. Develop fidelity of implementation instrument.	Draft Completed (Yr 1); Revised (Yr 2)
3. Develop learning modules for DBI-TLC.	Draft Completed (Yr 1); Revised (Yr 2)
4. Develop teacher knowledge and skills assessment.	Draft Completed (Yr 1); Revised (Yr 2)
5. Host Summer Institute for implementers.	Completed (Yr 1)
6. Plan collaborative support component of PD system.	Completed (Yr 2)
7. Develop and collect data on collaborative support activities.	In progress (Yr 2)
8. Make final revisions to activities.	In progress (Yr 2)
9. Conduct feasibility test of DBI-TLC, then revise DBI-TLC.	In progress (Yr 2)

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1. **Develop and revise the DBI protocol (Manual with processes and tools).** To accomplish this goal, we continued to revise and add to the DBI manual and tool-kits, including the addition of updated administration directions and scoring rules for CBM-W. While we had solicited detailed feedback from our experts (Graham, Deno, and Fuchs) and school partners (Marston, Tanner-Jones, and Conley) on the content of the manual in Year 1 of the project, we wanted additional feedback from our teacher participants. First, we asked teachers specific questions during the week-long Summer Institute (described below) and typed their responses into a template. A note-taker was assigned each day to capture teacher feedback during large group meetings. In addition, each team member had an Excel template with the questions that were posed for small-group discussion and typed teacher comments during these discussions. Questions were oriented to each module. For instance, questions for Module 1 included:

    - After today's introduction, what is your initial impression of the DBI process? What are your current questions about next steps?
    - In what ways could the DBI process be easily integrated into your teaching practice? What works well?
    - Conversely, in what ways might the DBI process be problematic for you?
    - From your educational practice, what situational factors might we need to know about with respect to this content?
    - What aspects of the DBI process (e.g. overall purpose, three areas of writing instruction) require additional clarification, instruction, or practice?
    - If additional clarification or instruction would be helpful, what would you like better/instead/or more of?
    - In today's session, which parts (lecture or activities) were most helpful to understanding the content? Which parts were unnecessary or confusing?

    After the Summer Institute, responses to all questions were combined into a master template. In addition, a brief survey was administered to the teachers via Qualtrics following the Summer Institute to capture their feedback in a way that provided more anonymity. Questions for the follow-up survey included:

    - What are your current questions about next steps for implementation?
    - Reflecting on the week, overall what were some of the best features of the PD we provided?
    - What were some features of the PD that could be improved? How?

    All sources of qualitative data were entered into the master excel template and coded using the following descriptors:

    - source
    - site
    - topic
    - question/issue
    - idea (from teacher discussion)

- possible area of development/refinement (from DBI team)
- priority
- decision/solution
- assigned to

The PIs reviewed all feedback and determined whether it should be addressed and the level of priority, based on the frequency it came up and level of importance or concern that teachers expressed related to the issue. Then, tasks were assigned to research team members to execute revisions.

For some of the issues that teachers had raised, the most direct way to address their comments was to create a document with frequently asked questions. For other issues, revisions were made to tools and learning modules. Finally, some of the issues were flagged as issues to be addressed in the "collaborative support" component of DBI-TLC.

Overall, data captured during the Summer Institute were extremely valuable in the revision process.

2. **Develop fidelity of implementation instrument.** We created several fidelity of implementation tools in Year 1 and revised them in Year 2. Specifically, we revised the Accuracy of Implementation Rating Scales (AIRS) forms that were created in Year 1 for implementation of each CBM-W measure (Word Dictation, Picture Word, and Story Prompt). This was in response to teacher feedback as well as slight revisions to the administration directions. Second, we created implementation fidelity checklists for the intervention components. A checklist with broad categories was created to assess Writing Intervention Plan (WIP) implementation for any activities that the teacher might have included (see portion below). In addition, a fidelity checklist was created to accompany each mini lesson in the writing toolkit. For example, for the mini lesson on word building, the accompanying checklist included specific procedures that the teacher should follow to implement that instruction. An implementation checklist was also created for decision-making, and included components that address teacher's attention to the decision making process for students. Fidelity checklists are provided in Appendices A-E.

   We are currently using these checklists for our "collaborative support" component (described below): coaches use them to observe the teachers' implementation of each step of the DBI process, and teachers are encouraged to use them for self-checks.

3. **Develop learning modules for DBI-TLC.** The learning module development was a collaborative project between the Minnesota and Missouri grant project teams. The learning modules for DBI-TLC were developed to align with the objectives in the DBI manual, and to focus on the 8 steps of the DBI process. We designed a framework, master plan, and templates for each PD module; then, content for each module was developed, along with corresponding material, by subsets of the research team. Originally, we developed 4 training Modules to be presented during our week long Summer Institute. These Modules were: Module 1- Using DBI; Module 2 – Assessing Student Progress using CBM; Module 3 –

Selecting and Implementing Appropriate Writing Interventions; and Module 4 – Making Data-Based Decisions.

Based on feedback we received from the Summer Institute (described below), we made revisions to the original learning modules and accompanying material. The main changes based on participant feedback and our team's reflection of the training included changing the order of presenting the learning modules to better reflect the 8 steps of DBI, condensing them into 3 learning modules, and breaking the module content into 3 single day trainings for teachers. By making these changes for our Feasibility Study (Year 2), we were able to provide teacher training in smaller chunks and a timelier manner.

In the revised set of learning modules, during Module 1, teachers learn about the first two steps of the DBI process (Establishing a present level of writing performance and Setting a long-term goal) and then have the opportunity to practice these skills with coaching support prior to the second training. The Module 2 training (Implement high-quality instruction with fidelity and Monitor progress) is designed to be completed with teachers a few weeks after the initial training. The Module 3 training (Use decision rules, Generate hypotheses about student progress, make instructional changes if needed, and continue the process) is then given to teachers during a final training.

Learning Modules will undergo one more round of revisions following completion of the Feasibility Study, at the end of Year 2. Current versions of Modules 1-3 are provided in Appendices F-H.

4. **Develop teacher knowledge and skills assessment.** A pre/post-assessment was developed to assess teachers' knowledge and skills of the DBI process. Questions were accessed from quizzes and assessments used in previous courses taught at the University level, as well as a series of questions based on reading CBM that were tailored to reflect the administration and scoring of writing CBM. The pre/post-assessment contained 45 items. Questions were primarily multiple-choice; however, a rank order question was included along with questions requiring participants to score a mock selection of Word Diction, Picture Word, and Story Prompt probes. On the pre-assessment only, a series of demographic questions preceded the actual assessment. Both assessments were prepared using Qualtrics, an electronic survey platform.

   Participants completed the assessment anonymously during time provided at the beginning and close of the Summer Institute (July/August 2014). Anonymity was maintained by asking teachers to create a 4-digit unique identifier that they entered in response to the first question on both the pre and post-assessment. At the end of each assessment, an open-ended question asked participants to write down anything else they knew about DBI and CBM.

   A review of teachers' results was conducted after the Summer Institute. Questions that supported scoring logic were scored in Qualtrics, while the rank order and CBM scoring questions were manually coded. Further, percentage of change was calculated from pre to post-assessment for all items. This information was put into a table to compare percentage correct at pre and post-assessment, percentage of change, and change in sample size.

Questions that were answered correctly by 100% of teachers on both assessments, along with questions that were answered correctly by 94% (all but one person) or 100% on pre-assessment and 94% or 100% on post-assessment, and questions that had low scores on post-assessment or decreased scores on post-assessment, were reworked and reviewed for rigor. These questions were either retained, eliminated, or revised. Suggestions for revised questions were reviewed by both the MO and MN teams. While the rank order question was retained, the scoring questions were reworked as multiple choice questions. These revisions were used to recreate both assessments for the feasibility study, and were similarly created and scored in Qualtrics.

Initial results from the Summer Institute reveal that the percent of change from pre to post-assessment ranged from a decrease of 18%, to an increase of 65% across the assessments. A paired samples *t*-test demonstrated that there were statistically significant differences between teachers' overall pre and post-assessment scores (t (16) = – 4.871, p = .000; *Pre-Assessment*: M = 31.71, SD = 5.966, Range = 21-41; *Post-Assessment*: M = 39.88, SD = 4.973, Range = 28-45), suggesting that teachers' participation in the week-long Summer Institute professional development series resulted in their increased knowledge and skills related to components of the DBI process.

5. **Host Summer Institute for implementers.** A five-day DBI-TLC Summer Institute was hosted by the Missouri team July 28-August 1 with 9 participants and the Minnesota team August 4-8 with 8 participants (see Table 2 for demographic information). The schedule of information presented and practiced follows:

Day 1: Introduction to the project

- Pre-assessment
- Learning Module 1: Data-Based Instruction (DBI) Overview
- Learning Module 2: Curriculum-Based Measurement (CBM) Overview and Administration

Day 2

- Learning Module 2 continued: CBM Scoring and Graphing
- Learning Module 3: Writing Intervention Introduction, Research, and Writing Instructional Plan

Day 3

- Learning Module 3 continued: Writing Intervention, Selecting and Implementing high-quality, research-based instruction for beginning writers

Day 4

- Learning Module 4: Data-Based Decision Making

Day 5: Putting It All Together

- Feedback from participants
- Post-assessment

Each day included multiple opportunities to gather feedback regarding the information presented.

Table 2
Summer Institute Teacher Demographics

DBI-TLC Summer Institute – July/August 2014				
	Minnesota (*n*=8)		Missouri (*n*=9)	
	n	%	n	%
Sex (female)				
Female	8	100	9	100
Race				
Black/African American	1	13	---	---
Native American	1	13	---	---
White/European American	6	75	8	89
Age				
20-29	---	---	1	11
30-39	1	13	2	22
40-49	3	38	2	22
50-59	4	50	4	44
Highest Degree				
Bachelor's	3	38	1	11
Master's	1	13	2	22
Master's + Coursework	4	50	4	44
Ed.S.	---	---	1	11
Ph.D.	---	---	1	11
Current Job Title				
Special Education Teacher	7	88	3	33
General Education Teacher	---	---	1	11
ESL Teacher	---	---	1	11
Other	1*	13	4**	44
	M	*Range*	*M*	*Range*
Years in current position	7.31	(6mo-22y)	7.22	(3-19y)
Years teaching elementary	13.75	(3-31y)	13.67	(0-26y)
Years teaching special ed.	9.13	(0-22y)	10.11	(0-30y)

Note: mo = months; y = years
* This participant from MN identified her current job as reading.
** Participants in MO who identified themselves as an administrator (Educational Diagnostician/Process Coordinator, Elementary Language Arts Coordinator, Supervisor, Director), detailed their primary responsibilities to include managing evaluation, coordinating meetings, conducting observations, coordinating intervention and progress monitoring, providing professional development, and overseeing special services.

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- All participants in MN primarily possess certification in elementary (grades 1-6) and special education (e.g., LD, EBD, grades 1-6 or Pre-K-12). One participant possessed certification in Reading.
- Four of 7 responses indicated that they had not received any training in CBM. Two reported coursework or hours in data-based decision making. One participant reported training in AVID.

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- All participants in MO primarily possess certification in elementary, early childhood, and/or special education (e.g., mild/moderate disabilities, LD, ID, cross categorical) in grades K-6, some possess certification in these same areas through grade 12. One participant also possessed National Board Certification and one participant was certified as a school psychologist.
- Of 7 participant responses, 2 participants have not received any training in CBM. The others have received some, albeit little training in writing, data-based decision making, and AIMSweb.

6. **Plan collaborative support component of PD system.** Collaborative support components of the DBI-TLC PD system were based on external and internal feedback that resulted in a guiding document of coaching principles for DBI-TLC. External feedback consisted of (a) feedback from teacher attendees of the Summer Institute, (b) a fall consultation with professional development expert, Stephanie Al Otaiba, and (c) a fall focus group with teacher attendees of the Summer Institute. Internal feedback included (d) a literature review on coaching and professional development conducted by MN and MO graduate assistants, and (e) MN and MO team discussions. External and internal feedback facilitated development of (f) the DBI-TLC principles of coaching, developed in the fall of 2014.

   Feedback from Summer Institute. Feedback from the Summer Institute was first recorded by graduate assistants using note-taking templates during Summer Institutes in MN and MO. Notes reflected teachers' responses to module content, collaborative support ideas, and the DBI process that were elicited during formal and informal discussions.

   These notes were then compiled, categorized, and prioritized within an Excel document designed to indicate areas in need of further development and refinement within DBI-TLC (as described above). This document indicated whether issues raised were better addressed by adding extra guidance to future trainings or materials, or if they required ongoing, intensive support. Issues requiring ongoing support were used to inform the nature of DBI-specific content addressed in collaborative support components of the PD system. Furthermore, ideas regarding *how* this content should be addressed were also identified. For example, possible professional development ideas included booster sessions, professional learning communities, or online assistance to address issues raised.

   Fall consultation with PD expert, Stephanie Al Otaiba. The second piece of external feedback regarding collaborative support components came from a Skype consultation with Dr. Stephanie Al Otaiba from Southern Methodist University. Dr. Stephanie Al Otaiba has expertise in teacher professional development and provided guidance regarding feedback from the Summer Institute (described above) to DBI PIs and project coordinators. This consultation occurred early in the Fall of 2014.

   During the consultation, we provided Dr. Al Otaiba an overview of the Summer Institute and teacher feedback related to collaborative support. In addition, we shared our plans for future revisions based on teacher feedback and how we planned to develop our collaborative support component during fall 2014. Dr. Al Otaiba provided useful suggestions related to the inclusion of digital PD enhancements including teacher-coach Skype sessions, video-recorded lessons, and online meetings. She encouraged us to consider whether we have the infrastructure to support online learning and provided several resources for further direction. An important highlight from this conversation included the suggestion to keep the decision-making component as the framework of our collaborative support and to consider other pieces as supporting details.

   Fall focus group. The third piece of external feedback regarding collaborative support components came from a focus group conducted in mid fall 2014. During this focus group,

teachers who attended the Summer Institute were invited to offer feedback on their implementation of DBI thus far. This information would inform the development of our collaborative support components by identifying how to make the DBI process more feasible/reasonable. Teachers provided updates of their implementation and responded to questions about the timing and necessity of content delivery, the level and type of coaching they preferred, and any additional insights based on their experiences.

Results of the focus groups for MN and MO indicated that MN teachers struggled to begin implementing DBI components whereas MO teachers had identified assessments for students and began collecting baseline data. In MN, teachers cited significant school-based barriers that limited their time and ability to begin DBI. For teachers in MN and MO, it was helpful to review how to access DBI materials from Google Drive and how to implement CBM-W assessments with fidelity.

Feedback from fall focus groups indicated that future support should occur immediately after teachers complete trainings to take advantage of teacher interest and momentum in DBI implementation. Furthermore, it was identified that teachers should be provided with hard copies of materials to limit time constraints related to their production and assemblage.

Coaching Literature Review. A literature review on coaching practices was conducted by MN and MO graduate assistants to inform development of DBI collaborative support components (see Appendix I for further description). The purpose of the literature review was to determine key characteristics of coaching within the academic literature related to data-based decision making. The findings from these articles guided the creation of coaching protocols and data collection methods.

Twelve studies met inclusion criteria for this literature review. Results of this review indicate that coaching is supported within the research as a viable practice, but lacks a consistent definition within the literature. Despite this, several common components were identified, including: non-evaluative assessment linked to constructive feedback, some form of accountability, supportive coach-teacher relationships, and individualized guidance or instruction on aspects of the intervention/skill being developed. Feedback was delivered through various means, but timely and on-going feedback appeared to be instrumental. Accountability was mainly achieved through goal setting, either by the experimenter or the teacher. The coach-teacher relationship was a central focus of much of the literature, but how these relationships were developed and/or evaluated was absent from most studies. Coaches modeled the target behavior, used guided practice, and observed teacher implementation, but how the coach decided which strategy to use was absent from most studies.

Based on these findings, graduate assistants synthesized a list of recommendations (discussed below) and presented them during team meetings for additional feedback and discussion.

Team discussions. Team discussions focused on coaching recommendations from the literature review as they related to data-based decision making and the DBI process. The purpose of team discussions was to identify how to include evidence-based coaching practices within DBI-TLC and to develop principles of coaching specific to the DBI process.

Based on the literature review described above, the following recommendations were synthesized and presented to teams:

General Recommendations:

- Teachers should set goals.
- Coaches should serve in non-evaluative roles.
- Differentiate coaching by teacher needs with a "tiered" coaching approach.
- Include a model, observe, feedback process (e.g., I do, We do, You do).
- Operationally define coaching as it relates to DBI.
- Offer initial training in small group and support groups.
- Connect to school-level outcomes, initiatives, and material.
- Seek out administrator support for implementation.
- Teach both pedagogical knowledge (the how) and content knowledge (the what)

Pre- Coaching Meeting

- Coach reviews notes from last meeting and reviews the teacher's goal(s)
- Prepares materials needed for meeting
- Confirms meeting in advance

During - Coaching Meeting

- Review teacher's goals for implementing DBI in his/her classroom
- Teacher shares self-reflections on progress
- Start with praise
- Look for one area to improve, set a goal, and make a plan
- Maintain focus on content and goals
- Incorporate fidelity checks for teacher/coach based on goals.

After - Coaching Meeting

- Record coaching actions in coaching log
- Record teaching actions in teaching log.

Principles of coaching. After discussion of coaching recommendations provided by the literature review in relation to data-based decision-making, we established a final list of DBI-TLC guiding coaching principles. The purpose of this list was to identify evidence-based coaching practices that would be effective within the DBI-TLC framework. In turn, the list would inform development of DBI-TLC coaching protocols incorporated across coaching sessions during the Feasibility Study.

We began by developing a general definition of coaching as it relates to DBI-TLC:

Coaching is a non-evaluative, ongoing, dynamic, and interactive process in which one individual (with training in DBI) observes, models, and provides feedback to another individual targeting the delivery of an intervention or practice, while encouraging the implementer to reflect on and gradually improve his/her practice and confidence in implementing the intervention or implementing the practice

The following five coaching principles were then identified:

1. *Coaching process should be teacher-oriented.* This principle focuses on building non-evaluative relationships that recognize the challenges associated with implementing DBI and fosters differentiated coaching approaches based on individual teacher needs.

2. *Coaching process should focus on building mastery.* This principle focuses on fidelity of implementation and building fluency with procedures. This is completed through teacher observations and the provision of modeling, guided practice, and constructive, timely feedback.

3. *Coaching process should be specific; desired coaching behaviors should be observable and measurable.* This principle focuses on a well-specified approach to coaching through standardized coaching agendas that include a consistent questioning framework to address teacher goals.

4. *Coaching process should emphasize alignment with existing curriculum and instructional practices.* This principle emphasizes that DBI-TLC is a process that can be used with existing programs or curricula, and that materials can be used across curricular areas.

5. *Consider long-term sustainability of implementation.* This principle focuses on future practices that involve sustainability of DBI-TLC including establishing administrative support, identifying how teachers can train teachers, and a continual emphasis on why the process is important and adaptable.

These five principles were embedded within standardized agendas and protocols used to facilitate coaching sessions across teachers (described below). Within coaching protocols, standardized coaching questions that directly address each principle were included with citations for corresponding principles (CP 1-5).

7. **Develop and collect data on collaborative support activities.** Development and data collection for collaborative support activities entailed three major development activities: (1) development and implementation of a Fall Trial of coaching activities, (2) development of the Coaching Cycle, and (3) development and implementation of a Coaches' Institute.

Fall Trial. After reviewing feedback from the Year 1 Summer Institute (described above) and determining that our Collaborative Support activities would include a coaching model, we decided that it would be helpful to conduct a Fall Trial of possible coaching activities to better understand teachers' needs. The Fall Trial was conducted simultaneously to the Coaching literature review and team discussions described above, and contributed to our development of the Principles of Coaching.

Participants. Following the Fall Focus group (described above), teachers from the Summer Institute from both sites were invited to receive a limited amount of coaching to get them started with the DBI process. One teacher from MN, and two teachers from MO, agreed to participate in this Fall Trial. In MN, the PI served as the coach, and in MO, the PI and one project coordinator served as coaches, for these three teachers.

Measures and Materials. During the Fall Trial, we developed a "Status of Implementation" checklist to help guide coaching activities. This checklist included each step of the DBI process, the materials needed to implement it, and a place to indicate whether the teacher had started or not started the step and whether she needed help.

In addition, each Coach maintained a log of coaching activities. The purpose of this log was to identify the major issues that arose for teachers, the type of support they needed, and what it took to resolve these issues, in order to be able to anticipate the types of support that future teachers might need. The log was an excel spreadsheet with the following information entered for each coaching session: date, time allocated (in min), issue addressed, support mode (e.g., phone, email, face-to-face), outcome, & notes (including whether issues are initial, ongoing).

Procedures. In October-November of Year 2, research staff serving as Coaches met with the three teachers on an as-needed basis. Meetings were loosely structured and designed to address the specific individual needs of the teachers, based on where they were in the DBI process (as indicated by the Status of Implementation Checklist.

Results. Meetings were conducted between 1 and 4 times for each teacher and lasted between 45 and 90 min. While each teacher had made some attempt to begin administering CBM assessments, it was clear that none of them had gotten far in the process before coaching began. Teachers needed support in the following areas:

(a) Accessing the DBI tools from the Google drive that was provided to them over the summer.
(b) Reviewing CBM scoring procedures.
(c) Using the graphing tool.
(d) Setting baseline and long-term goals.

Teachers also expressed needs for support in developing instructional plans and monitoring student progress; however, there was not time to address these issues during the Fall Trial. Those who wished to continue receiving Coaching support were invited to participate in the Spring Feasibility study.

Development of Coaching Cycle. Using the Principles of Coaching described in Item #6 above, and results from the Fall Trial, we developed a Coaching Cycle that would be implemented across three Modules that would extend from January through May of Year 2. Details about the specific Modules and Coaching activities are provided in Table 3. For each module, the following Coaching Cycle was developed. Face-to-face Coaching was to be

provided at least twice between each Workshop, with "Virtual Check-Ins" provided in weeks when face-to-face coaching did not take place.

(a) Workshop (Modules 1-3; each delivered in a full-day face-to-face session).
(b) Performance Assessment (designed to determine whether teacher had the necessary skills to implement the DBI steps learned in the workshop).
(c) Coaching Protocol A (Coach checks teacher's Performance Assessment, provides feedback, addresses questions and provides any needed additional support, and makes plans/sets goals for teacher to begin implementing DBI steps using Plans & Goals sheet—see Appendix J).
(d) Observation (Coach observes teacher's implementation of the DBI steps learned in the workshop using the Accuracy of Implementation Checklist).
(e) Coaching Protocol B (Coach would provide feedback based on observation, address questions and provide any needed additional support, and make plans/set goals for teacher to continue implementing DBI steps using Plans & Goals sheet).
(f) Ongoing Coaching Protocol (Coach continues to provide face-to-face support as needed using Plans & Goals sheet).

Table 3
Modules and Coaching Activities

Module	DBI Steps	Coaching Activities	Materials
1. DBI Overview; Establishing Baseline and Setting a Long-Term Goal.	1-2	Coaching A: Review Performance Assessment 1; work with teacher to set goals for implementing Steps 1-2.	• Performance Assessment 1: Score each type of CBM-W prompt with > 85% accuracy. • Plans & Goals Sheet
		Observation 1: Observe teacher administering CBM assessment	• AIRS: Administering CBM
		Coaching B: Provide feedback and support for Steps 1-2.	• Plans & Goals Sheet
2. Implementing High Quality Writing Instruction with Fidelity; Monitoring Student Progress toward the Goal	3-4	Coaching A: Review Performance Assessment 2; work with teacher to set goals for implementing Steps 3-4.	• Performance Assessment 2: Develop a Writing Instructional Plan and provide a rationale. • Plans & Goals Sheet
		Observation 2: Observe teacher implementing instruction	• AIRS: Writing Intervention
		Coaching B; ongoing coaching: Provide feedback/support for Steps 3-4.	• Plans & Goals Sheet
3. Data-based Decision Making	5-8	Coaching A: Review Performance Assessment 3; work with teacher to set goals for implementing Steps 5-8.	• Performance Assessment 3: Use graphed performance to make data-based decisions. • Plans & Goals Sheet
		Observation 3: Compare CIP to graphed data	• AIRS: Decision-making
		Coaching B; ongoing coaching: Provide feedback/support for Steps 5-8.	• Plans & Goals Sheet

Coaches' Institute

After the thorough literature review conducted by our graduate students and the identification of essential coaching elements, we prepared our team to conduct high quality coaching practices. We scheduled a 3-hour "Coaches' Institute" with our project coordinators and graduate research assistants in both MN and MO. During this institute, we discussed the guiding principles of coaching, and then provided examples and non-examples of these practices. We provided a great amount of time for discussion and allowed participants to ask questions and provide examples of their own. We demonstrated non-appropriate coaching techniques through a video example, and then dissected the variety of coaching mistakes made. We asked participants to role play coaching using scenarios. Finally, we went through our initial coaching protocols, discussing each step in the agenda and prompting for questions. The Training Power Point can be found in Appendix K.

8. **Make final revisions to activities.** Revisions to Coaching Activities are occurring on an on-going basis. We are currently implementing our Feasibility Study (described below). During this study, we hold weekly meetings with our coaches to hear how things are going and problem-solve any issues that have arisen. During these meetings, we are keeping track of ways to improve the coaching process and materials. At the end of the Feasibility Study, we will hold a focus group session and implement an online survey with participating teachers, during which time we will gather additional feedback about their impressions of what works well or needs to be improved with respect to coaching.

9. **Conduct feasibility test of DBI-TLC, then revise DBI-TLC.**

Participants. Participants in the feasibility study include special education and English Language Learner teachers in MN and MO. At the beginning of the study, there were 8 new teachers in MN and 7 new teachers in MO (an additional 3 teachers from the MO Summer Institute also opted to continue participating).

One teacher in each site dropped at the beginning of the study, both due to family emergencies that prevented them from attending the first module. They both opted not to continue participating. It should also be noted that two MN teachers decided to discontinue implementing DBI components about halfway through the project, citing due process responsibilities taking precedence over study activities. These teachers agreed to still provide feedback regarding the feasibility of DBI-TLC, which we thought would be particularly important given that they found the procedures to be overwhelming in addition to other job responsibilities.

Demographic information for participating teachers in both sites is provided in Table 4.

Table 4
Feasibility Study Teacher Demographics

DBI-TLC Feasibility Study – Winter 2015				
	Minnesota (*n*=8)		Missouri (*n*=7)	
	n	%	*n*	%
Sex (female)	8	100	7	100
Female				
Race				
White/European American	7	88	7	100
I prefer not to respond	1	13	---	---
Age				
20-29	---	---	4	57
30-39	3	38	1	14
40-49	5	63	2	29
Highest Degree				
Bachelor's	1	13	1	14
Master's	5	63	3	43
Master's + Coursework	2	25	3	43
Current Job Title				
Special Education Teacher	8	100	4	57
ESL Teacher	---	---	2	29
Other	---	---	1*	14
	M	*Range*	*M*	*Range*
Years in current position	10.25	(2-22y)	5.57	(<1-19y)
Years teaching elementary	15.50	(8-24y)	6.43	(2-13y)
Years teaching special ed.	14.00	(2-24y)	4.29	(0-19y)

Note: y = years
* This participant from MO identified her current job as reading recovery/instructional coach, and described her current job responsibilities as meeting with teachers to help them plan, reflect, and set goals, and meeting with 1st grade students one-on-one to support reading and writing needs.

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- All participants in MN primarily possess certification in elementary (grades 1-6) and special education (e.g., LD, EBD, ASD, MMI, MMH, M/M, DCD grades 1-6 or Pre-K-12). One participant possessed certification in English grades 7-12, and one participant possessed certification in social studies grades 6-8.
- Four of 8 responses provided no response to the question asking if they had previously received training in or taken coursework in CBM. The remaining participants identified that they have received between 1 and 12 hours of PD or coursework in data-based decision making.

MISSOURI

- All participants in MO primarily possess certification in elementary, early childhood, and/or special education (e.g., mild/moderate disabilities, cross-categorical, LD, ID, cross categorical) in grades K-6, some possess certification in these same areas through grade 12. Two participants possessed certification in ESL, ESOL, Spanish, French, and Adult Education and Literacy.
- Of 7 participant responses, 2 participants provided no response to the question asking if they had previously received training in or taken coursework in CBM. One reported no PD except for what was completed in collaboration or in PLCs. One reported limited training in AIMSweb and one reported a brief training in CBM. The others reports training specific to their teaching assignments.

Measures & Materials. Measures and materials for the feasibility study include the following:

Teacher's data logs. All participating teachers were asked to maintain a data log of time spent conducting study activities, including number of minutes spent preparing, administering, and scoring CBM; graphing and examining CBM data; developing hypotheses; implementing intervention; and conducting the entire DBI process. Teachers were also encouraged to provide any additional information regarding the feasibility of the DBI process.

Coaches logs. All coaches were asked to maintain of log of time spent conducting coaching activities, including number of minutes spent in all coaching activities, the issues addressed in each coaching session, the mode of support (e.g., face-to-face, email, etc.), the outcome of the coaching session, and any additional notes that could be used to facilitate weekly team discussions of coaching activities.

Changes in Instructional Plans (CIPs). Following Module 3, in which teachers learn to make data-based instructional decisions, teachers were asked to complete CIPs each time they make instructional changes. The CIPs ask teachers to record the following:

- *Current writing instruction & data to support change*
- *Type of change to be made (hypothesis)*
- *With whom the change will be implemented*
- *Student to teacher ratio*
- *Frequency*
- *Duration*
- *Description of what's needed*
- *Set new goal (as needed) and indicate review date*

The CIP will be used in conjunction with student graphs to evaluate the timeliness and appropriateness of teacher decisions.

Procedures. The Feasibility Study is being conducted as follows:

December 2014. We recruited teachers from MN and MO with the help of our school partners in Minneapolis and Columbia Public Schools.

January-April 2015. We implemented our Coaches' Institute, and invited teachers to the first workshop. The invitation included a request to complete the Knowledge and Skills pre-assessment via Qualtrics. The Module 1 workshop took place in each site, followed by Module 1 Coaching Sessions A and B. Within two to three weeks, teachers were invited back for the Module 2 workshop, followed by Module 2 Coaching Sessions A and B, and ongoing coaching (face-to-face and/or virtual) until the final Module 3 workshop (approximately 6-8 weeks after Module 2 workshop). Teachers were asked to continue implementing DBI with 2-3 students for a total of 12 weeks. Coaches will be audio-recording Coaching Session A of Module 3, to be used for development of a Coaching Fidelity of Implementation measure.

May 2015. We will hold a focus group, at which time we will ask for teachers' feedback regarding the feasibility and usability of the DBI process, tools, learning modules, and coaching support. Teachers will also complete the post-assessment, a teacher efficacy measure (which we are trying out for use in our Year 3 Pilot Study), and an anonymous survey to give additional feedback about coaching.

Data Analysis. Data analysis will be conducted to (a) determine the feasibility and usability of DBI-TLC, and (b) develop coding systems and measures for the pilot study.

Determining feasibility/usability. To determine the feasibility and usability of DBI-TLC, we will examine whether teachers made significant pre- to post-assessment growth using dependent samples *t*-tests on the Knowledge and Skills assessment; examine fidelity checks conducted by the DBI coaches to determine whether teachers were able to implement DBI steps with fidelity; and analyze qualitative feedback data in a similar manner as the approach described under item #1 above. Specifically, all feedback items will be entered into an excel spreadsheet, categorized based on common themes, and discussed by team members to identify and prioritize specific revisions needed for DBI-TLC.

Developing coding systems. For the pilot study, the following measures will be needed to assess elements of our theory of change (shown in Figure 1), and will be developed and/or refined based on Feasibility Study activities, as follows:

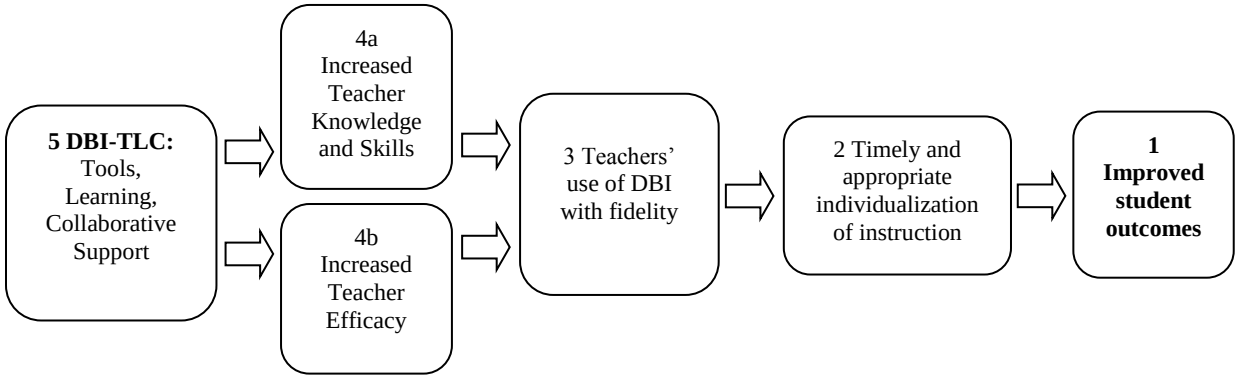


Figure 1. ***Theory of Change.***

- *Teachers' Knowledge and Skills of DBI*: We will refine the Knowledge and Skills assessment by examining the numbers of questions teachers got correct at pre-test vs. posttest and revising/refining items as needed.
- *Teachers' Self-Efficacy*: We will examine teachers' responses to the Self-Efficacy measure, and discuss our findings with Content Expert Steve Graham. Based on the data and Dr. Graham's input, we will revise/refine this measure as needed.
- *Teachers' Fidelity of Implementation of DBI*: We will refine the AIRS checklists based on feedback from Coaches, who conducted fidelity observations with the current drafted checklists.
- *Teachers' Timeliness and Appropriateness of Instructional Decisions*: We will examine the timeliness and appropriateness of decisions by comparing CIPs to

student graphs, and developing a coding system to record these dimensions of decision-making.

- *Coaching Fidelity*: We will review audio-recordings of Coaching sessions to develop a Coaching Fidelity of Implementation measure.

C. What opportunities for training and professional development has the project provided? Thus far, GRAs on the project have received training and experience in the following areas:

- CBM-Writing administration, scoring, and interrater agreement procedures.
- Development of tools related to writing assessment and intervention.
- Cross-site collaboration.
- Synthesizing literature and conducting article reviews.
- Delivering DBI content through learning modules.
- Developing, conducting, and evaluating coaching activities.
- Discussion regarding IES goal structures and grant implementation.

D. How have the results been disseminated to communities of interest? We have disseminated aspects of the project (mainly, materials that have been developed) to the following communities of interest in the following ways:

Information/Materials Disseminated	**Communities of Interest**
1. Initial findings of writing intervention literature review	Research community (Pacific Coast Research Conference, Society for Scientific Studies of Reading), Practitioner community (Council for Exceptional Children, Council for Learning Disabilities)
2. Assessment and intervention materials	Teacher preparation community (Instructors at MN and MO institutions, as well as colleagues at other national institutions); State agencies (MO Council for Exceptional Children)
3. Training on use of DBI components	Practitioner community (General educators, school administrators, school psychologists)

E. What do you plan to do during the next reporting period to accomplish the goals? Table 5 provides a summary of activities to be completed in the next reporting period.

Table 5
Project Activities and Timeline

Project Activity (As listed in Performance Agreement)	**To be completed by:**
Complete feasibility test of DBI-TLC, then revise DBI-TLC.	
• Complete Module 3 with coaching, hold focus group	May 2015
• Analyze feasibility data and plan revisions	May-June 2015
• Execute revisions	June-July 2015
Conduct small randomized pilot study of DBI-TLC.	
• Recruit teachers	May 2015
• Plan study (face-to-face PI meeting)	June 2015
• Finalize measures and materials	July 2015
• Implement revised Learning Modules	Aug-Dec 2015
• Teachers implement DBI and receive ongoing coaching	Sep 2015-Feb 2016
Collect pre and posttest data from teachers and students along with fidelity data.	Sep 2015-March 2016
Analyze data and disseminate findings.	April-August 2016

PRODUCTS

A. Publications and Presentations

Jung, P. & McMaster, K. L. (2015, April). Promise of data-based instruction (DBI) for improving students' early writing performance. Multiple presentation sessions at the 2015 Council for Exceptional Children Convention, San Diego, CA.

Poch, A., Smith, A., Jung, P., McMaster, K. L., & Lemke, E. (2015, April). Establishing effective academic coaching: A review of the coaching literature. Poster presentation at the 2015 Council for Exceptional Children Convention, San Diego, CA.

Jung, P. & McMaster, K. L. (2015, March). Data-based instruction (DBI) in early writing. College of Education and Human Development Research Day, College of Education and Human Development, University of Minnesota, Minneapolis, MN.

Lembke, E., Moore, K., & Herriges, M. J. (2015, February). Early writing CBM tools: Administration, scoring and data utilization. Mini skills session at the annual convention of the National Association of School Psychologists, Orlando, FL.

McMaster, K. L., Kunkel, A., Shin, J., & Lembke, E. (2015, February). Interventions for beginning writers: A literature review. Presented at the 2015 Pacific Coast Research Conference, San Diego, CA.

Jung, P. & McMaster, K. L. (2015, February). Data-based instruction (DBI) in early writing. Presented at the 2015 Pacific Coast Research Conference, San Diego, CA.

Carlisle, A., & Poch, A. (October 2014). *Incremental administration across grades for word dictation CBM-W.* Poster Presentation at the 36th Annual Conference on Learning Disabilities, Philadelphia, Pennsylvania.

Lembke, E., & Poch, A., & Carlisle, A. (September 2014). *Data-based decision making in early writing: Getting students off to the 'write' start*. Presentation at the Missouri Council of Administrators of Special Education, Lake of the Ozarks, Missouri

Brandes, D., Poch, A., Jung, P-G., Shin, J., & Kunkel, A. (April 2014). *Gender differences in performance on curriculum-based measures of early writing*. Poster Presentation at the DLD Reception at the Council for Exceptional Children's Exposition, Philadelphia, Pennsylvania.

B. Website (currently under development at each site)

C. Other Products

DBI-TLS products include:

Tools for DBI

- DBI Manual
- Curriculum-Based Measurement Tools for Early Writing
    - Assessments (Word Dictation, Picture Word, Story Prompts)
    - Selecting CBM Tool
    - Graphing Tool
- Writing Intervention Tools
    - Selecting Research-Based Writing Interventions Tool
    - Literature Summary Tool
    - Writing Intervention Plans (WIPS--3 versions; Transcription, Text Generation, Self-Regulation)
    - Mini-Lessons (for Transcription and Text Generation)
- Decision-Making Tools
    - Decision-Making Rubric
    - Changes in Instruction Plans form
- Fidelity Tools (see Appendices A-E)

Learning Modules (see Appendices F-H)

- Module 1: DBI Overview; Establishing Baseline and Setting a Long-Term Goal
- Module 2: Implementing High Quality Writing Instruction with Fidelity and Monitoring Student Progress toward the Goal
- Module 3: Data-based Decision Making

Coaching Materials (see Appendices J-Q)

- Coaches' Institute training
- Plans & Goals Sheet (used during each coaching session)
- Module 1:
    - Coaching Protocols A & B
    - Performance Assessment 1: Score each type of CBM-W prompt with > 85% accuracy.
- Module 2:
    - Coaching Protocols A & B, ongoing
    - Performance Assessment 2: Develop a Writing Instructional Plan and provide a rationale.
- Module 3:
    - Coaching Protocols A & B, ongoing
    - Performance Assessment 3: Use graphed performance to make data-based decisions.

PARTICIPANTS AND OTHER COLLABORATING ORGANIZATIONS

A. What individuals have worked on the project?

Name	Project Role	Person mo.	Contribution	Funding Support	Foreign Country
Kristen McMaster	Co-Principal Investigator	1.8 (AY), 1 (Su)	Oversees all aspects of project; coordinates with Content Experts and School Partners	IES	NA
Erica Lembke	Co-PI	1.8 (AY), 1 (Su)	Oversees all aspects of project; coordinates with Content Experts and School Partners	IES	NA
Ted Christ	Co-Investigator	0.9 (AY), .45 (Su)	Provides leadership in assessment aspects of the project	IES	NA
Kim Moore	MO Project Coordinator	3.0	Helps coordinate project, develops materials, provides PD and coaching, contributes to report writing	IES	NA
Carol Garman	MO Project Coordinator	3.0	Helps coordinate project, develops materials, provides PD and coaching, contributes to report writing	IES	
Dana Brandes	MN Graduate Research Assistant	6.0	Helps coordinate project, develops materials, provides PD and coaching, contributes to report writing	IES	NA
Pyung-Gang Jung	MN Graduate Research Assistant	3.0	Completes literature review and development tasks, provides PD and coaching, contributes to report writing	IES	NA
Jaehyun Shin	MN Graduate Research Assistant	3.0	Completes literature review and development tasks, provides PD and coaching, contributes to report writing	IES	NA
Michael Herriges	MN Graduate Research Assistant	6.0	Completes literature review and development tasks, provides PD and coaching, contributes to report writing	IES	NA
Kyle Wagner	MN Graduate Research Assistant	3.0	Completes literature review and development tasks, provides PD and coaching, contributes to report writing		
Abby Carlisle	MO Graduate Research Assistant	3.0	Completes literature review and development tasks, provides PD and coaching, contributes to report writing	IES	NA
Apryl Poch	MO Graduate Research Assistant	3.0	Completes literature review and development tasks, provides PD and coaching, contributes to report writing	IES	NA
Alex Smith	MO Graduate Research Assistant	1.5	Completes literature review and development tasks, provides PD and coaching, contributes to report writing	IES	NA

What other organizations have been involved as partners?
Minneapolis Public Schools, MN and Columbia Public Schools, MO are serving as the district sites for all school-based activities described in this report.

Have other collaborators or contacts been involved? No.

IMPACT

A. Impact on the development of the principal disciplines of the project

The current project has the *potential* to impact the principal disciplines (writing, special education) in the following ways:

- Improve struggling writers' outcomes in writing.
- Improve teachers' administration, scoring, and use of CBM in writing.
- Improve teachers' knowledge, skills, and self-efficacy related to DBI.
- Facilitate teachers' implementation of DBI.
- Facilitate teachers' capacity to provide instruction targeting multiple dimensions of writing.
- Contribute to knowledge in the field regarding how to provide PD and support related to teachers' implementation of DBI in writing.

B. Impact on other disciplines (None at this time.)

C. Impact on the development of human resources

This project has provided opportunities for project coordinators, graduate research assistants and teachers to learn about theoretical models of writing, assessment and intervention for beginning writers, intervention design, research design, and to gain experience developing and piloting a variety of assessments and intervention protocols. In addition, project coordinators and graduate research assistants have learned to design and implement a coaching model to support teachers' use of evidence-based practices.

D. Impact on resources that form infrastructure

None at this time.

E. Impact on technology transfer

None at this time.

F. Impact on society

None at this time.

G. Dollar amount spent in foreign countries

NA

CHANGES/PROBLEMS

There are no changes or problems to report.

BUDGETARY INFORMATION

Below is a description of project expenses through 2/28/15.

Principal Investigator Kristen McMaster, Ph.D. (1.8 person months AY, 1 month summer)
Dr. McMaster has been collaborating with Dr. Lembke to oversee all major grant activities, including budget management; supervision of project coordinator and GRAs at their respective sites; study design and planning; development and revision of all intervention components; coordinating with School Partners for planning, recruiting, and conducting research in schools; obtaining feedback from School Partners, Content Experts, and DBI Implementers; conducting the Summer Institute, Teacher Workshops, and other teacher meetings; overseeing school-based study activities; data analysis and interpretation; writing reports; and disseminating findings.

Co-Investigator Theodore Christ, Ph.D. (.90 person months AY, .45 summer)
Dr. Christ participates in study design and planning,; development and revision of intervention components and Teacher Knowledge and Skills assessment, and meetings with School Partners, Content Experts, and DBI Implementers.

Project Coordinator Dana Brandes (6.0 person months)
The Project Coordinator is responsible for coordinating and carrying out ongoing project activities, including training and supervising GRAs; scheduling, coordinating, and attending all meetings; communicating regularly with co-PIs, GRAs, School Partners, and Content Experts; coordinating teacher workshops; coordinating school-based study activities; coordinating data coding, entry, and analysis; assist in writing reports and disseminating results.

Graduate Research Assistants (GRAs; 1.5 to 3.0 person months)
GRAs assist with study planning, development of materials, recruitment, participating in all research team meetings, data collection and coding/scoring/entry, and conducting classroom consultation visits and observations. GRAs attend weekly project meetings to provide PI and Co-I with updates and to engage in any problem solving needed to provide support to teachers.

Fringe Benefits
Fringe benefits are calculated at the University of Minnesota academic rate of 33.8% for PI and Co-I. Graduate student fringe benefits, including the Project Coordinator's, are calculated at 16.6% plus $17.84 per hour tuition remission.

Travel ($2,218)
Travel costs have been incurred for traveling to conference to present research related to the project.

Materials and Supplies ($1,887)
Materials have been purchased for development and teacher training activities.

Consultant/Professional Services ($8,225)
Payments to Content Experts (Al Otaiba, Deno, Fuchs, Graham), School Partners, and participating teachers

Subaward ($218,161.00)
Payments made to University of Missouri to date.

Other Expenses ($921)
Other expenses include printing and duplicating of materials for teacher training and support activities.

Changes to the Budget
None this year.